

Disproof of conjecture 00000003238:
a direct sum of nilpotent Jordan blocks has the closed unit disc as
spectrum, while every block has spectrum $\{0\}$

Jackmeson1

2026-10-06

1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Edge spectra: the block spectral structure of direct-sum operators. Conjecture: The edge theorem: the closure of the spectrum of a direct sum of operators is the union closure of block spectra, with limit points the accumulated edges of the family of summands. (direct sum block spectrum accumulated edge) (The Chinese text states the same.)

Reading. Let $(H_n)_{n \geq 0}$ be complex Hilbert spaces and (A_n) a uniformly bounded family of bounded operators $A_n : H_n \rightarrow H_n$. Their direct sum is the operator $\bigoplus_n A_n$ on the Hilbert direct sum $\bigoplus_n H_n$ (square-summable sequences $x = (x_n)$, $x_n \in H_n$) given by $(\bigoplus_n A_n)x = (A_n x_n)_n$. The conjecture's main clause is

$$(E) \quad \overline{\sigma\left(\bigoplus_n A_n\right)} = \overline{\bigcup_n \sigma(A_n)}.$$

The clause “with limit points the accumulated edges of the family of summands” is joined to (E) by “with” (Chinese: “and”); it uses the undefined term “accumulated edges” and is not needed: refuting (E) refutes the conjecture. We refute (E) for one explicit family of contractions ($\|A_n\| \leq 1$) on finite-dimensional spaces $H_n = \mathbb{C}^{n+1}$.

Readings not refuted. The phrases “union closure”, “limit points” and “family of summands” indicate an infinite family, which is what we refute. For a *finite* direct sum the identity (E) is true, and it is also true for families of *normal* operators (for normal A_n and λ at positive distance d from $\bigcup_n \sigma(A_n)$, the resolvents satisfy $\|(A_n - \lambda)^{-1}\| \leq 1/d$, so they assemble into a bounded inverse). Neither restriction appears in the text, which says “a direct sum of operators” with no normality hypothesis. We also only treat the Hilbert (ℓ^2) direct sum, which is the standard direct sum of Hilbert space operators.

2 Definitions (as formalized)

All conventions below are those of the Lean file.

- $\mathbb{N} = \{0, 1, 2, \dots\}$. Block n is $\mathbb{C}^{n+1}$ with the Euclidean norm (`Block n := EuclideanSpace C (Fin (n+1))`), coordinates indexed by $0, \dots, n$.
- The Hilbert direct sum of a family $E : \mathbb{N} \rightarrow \text{Type of complex inner product spaces}$ is Mathlib's `lp E 2`, the square-summable sections, with Mathlib's inner product structure

`lp.instInnerProductSpace` (Mathlib calls it the “Hilbert sum” of the family). For our blocks it is complete (a Hilbert space).

- `dsum A hA`: for $A_n : E_n \rightarrow E_n$ bounded with $\|A_n\| \leq 1$, the bounded operator on `lp E` 2 with $(\text{dsum } A \text{ hA } x)_n = A_n x_n$ (theorem `dsum_apply`); it maps the n -th summand by A_n (`dsum_single`), is the unique bounded operator acting blockwise as (A_n) (`dsum_unique`), and $\|\text{dsum } A \text{ hA}\| \leq 1$ (`norm_dsum_le`).
- Spectrum: Mathlib’s `spectrum C a` = $\{\mu \in \mathbb{C} : \mu \cdot 1 - a \text{ is not a unit}\}$, computed in the Banach algebra of bounded (continuous) linear operators. Being a unit there means having a bounded two-sided inverse.
- The shift J_n on $\mathbb{C}^{n+1}$: $(J_n v)_j = v_{j+1}$ for $j < n$ and $(J_n v)_n = 0$ (`shift n`). Equivalently $J_n e_0 = 0$, $J_n e_{j+1} = e_j$: a single nilpotent Jordan block of size $n + 1$.
- **EdgeTheorem**: the statement that (E) holds for every family $E : \mathbb{N} \rightarrow \text{Type}$ of complex Hilbert spaces and every family (A_n) with $\|A_n\| \leq 1$. Contractive families are uniformly bounded, so **EdgeTheorem** is a special case of the conjecture’s claim.

3 Proof

Lemma 1 (blocks). $\|J_n\| \leq 1$, $J_n^{n+1} = 0$, and $\sigma(J_n) = \{0\}$. Hence $\overline{\bigcup_n \sigma(J_n)} = \{0\}$.

Proof. $\|J_n v\|^2 = \sum_{j < n} |v_{j+1}|^2 \leq \|v\|^2$. By induction on k , $(J_n^k v)_j = v_{j+k}$ if $j + k \leq n$ and 0 otherwise, so $J_n^{n+1} = 0$. If $\mu \neq 0$, then $\mu \cdot 1 - J_n$ is the sum of the unit $\mu \cdot 1$ and the nilpotent $-J_n$, which commute, so it is a unit. For $\mu = 0$, $-J_n$ is nilpotent in a nontrivial ring, so it is not a unit. $\square$

Let $T = \bigoplus_n J_n$ (`T := dsum shift norm_shift_le`) on $\bigoplus_n \mathbb{C}^{n+1}$; $\|T\| \leq 1$.

Lemma 2 (test vectors). For $z \in \mathbb{C}$ let $v^{(n)} = (1, z, z^2, \dots, z^n) \in \mathbb{C}^{n+1}$. Then $(z \cdot 1 - J_n)v^{(n)} = z^{n+1}e_n$, and in the direct sum $(z \cdot 1 - T)\iota_n(v^{(n)}) = \iota_n(z^{n+1}e_n)$, where ι_n is the inclusion of the n -th summand.

Proof. For $j < n$ the j -th coordinate is $z \cdot z^j - z^{j+1} = 0$; the n -th is $z \cdot z^n - 0$. The second identity follows since $T\iota_n = \iota_n J_n$. $\square$

Theorem 3. $\sigma(T)$ is the closed unit disc. In particular $\frac{1}{2} \in \sigma(T)$.

Proof. $(\subseteq)$ $\sigma(T) \subseteq \{|z| \leq \|T\| \|1\|\}$ and $\|T\| \leq 1$, $\|1\| \leq 1$. $(\supseteq)$ Let $|z| < 1$ and suppose $z \cdot 1 - T$ had a bounded inverse S . Put $x_n = \iota_n(v^{(n)})$. Then $\|x_n\| = \|v^{(n)}\| \geq |v_0^{(n)}| = 1$, and by Lemma 2

$$1 \leq \|x_n\| = \|S(z \cdot 1 - T)x_n\| \leq \|S\| \|z^{n+1}e_n\| = \|S\| |z|^{n+1} \quad (n \geq 0).$$

Choose n with $|z|^n < 1/(\|S\| + 1)$; then $\|S\| |z|^{n+1} \leq \|S\| |z|^n < 1$, a contradiction. So the open disc lies in $\sigma(T)$; since $\sigma(T)$ is closed, so does its closure, the closed disc. $\square$

Theorem 4 (main). $\overline{\sigma(\bigoplus_n J_n)} \neq \overline{\bigcup_n \sigma(J_n)}$. Hence **EdgeTheorem** is false, and the conjecture’s clause (E) fails for the uniformly bounded family (J_n) .

Proof. By Lemma 1 the right side is $\{0\}$; by Theorem 3 the left side contains $\frac{1}{2} \neq 0$. $\square$

4 Lean formalization and axiom audit

File `lean/Conjecture3238/Basic.lean` (273 lines, only `import Mathlib`, `Lean` and `Mathlib v4.33.1`), namespace `C3238`. Main statements:

- `theorem not_edgeTheorem : ¬ EdgeTheorem`
- `theorem edge_theorem_counterexample : closure (spectrum C T) ≠ closure (⋃ n, spectrum C (shift n))`
- `theorem spectrum_T : spectrum C T = Metric.closedBall 0 1`
- `theorem spectrum_shift (n) : spectrum C (shift n) = {0}`
- `theorem closure_iUnion_spectrum_shift : closure (⋃ n, spectrum C (shift n)) = {0}`
- `theorem mem_spectrum_T {z} (hz : ||z|| < 1) : z ∈ spectrum C T`
- supporting: `dsum_apply`, `dsum_single`, `dsum_unique`, `norm_dsum_le`, `norm_shift_le`, `shift_pow_eq_zero`, `block_identity`, `sum_identity`.

Here `C` stands for $\mathbb{C}$ and `T := dsum shift norm_shift_le`.

Axiom audit. `#print` axioms for `not_edgeTheorem`, `edge_theorem_counterexample`, `spectrum_T`, `spectrum_shift` and `dsum_unique` reports exactly `[proptext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide`, and no added axiom.

Sources

- Wikipedia, *Spectrum (functional analysis)*, [https://en.wikipedia.org/wiki/Spectrum_\(functional_analysis\)](https://en.wikipedia.org/wiki/Spectrum_(functional_analysis)): “the spectrum of a bounded linear operator (or, more generally, an unbounded linear operator) is a generalisation of the set of eigenvalues of a matrix” and “More generally, by the bounded inverse theorem, T is not invertible if it is not bounded below”. (Background only; the Lean proof of $\frac{1}{2} \in \sigma(T)$ is self-contained.)
- Mathlib, `Mathlib/Analysis/InnerProductSpace/l2Space.lean` (definition of the Hilbert ℓ_2) and `Mathlib/Algebra/Algebra/Spectrum/Basic.lean` (`spectrum`).